Hybrid-MambaCD: Hybrid Mamba–CNN Network for Remote Sensing Image Change Detection With Region–Channel Attention Mechanism and Iterative Global–Local Feature Fusion

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***Abstract*— Mamba has gained significant attention for its outstanding long-range context modeling capability while maintaining linear complexity, compared with Transformer. In this article, a hybrid Mamba and convolutional neural network (CNN) architecture is proposed for remote sensing image change detection (RSICD), named Hybrid-MambaCD, which leverages the advantages of CNN for local detail information extraction and Mamba for global context information extraction, providing an efficient solution for RSICD tasks. First, the region–channel attention mechanism (RCAM) is designed to enhance the CNN features from both channel and region dimensions, enabling the network to focus more on change regions while suppressing interference from background areas. Second, an iterative global–local feature fusion (IGLFF) strategy is proposed, which performs an adaptive weighted fusion of global and local features across multiple scales in a progressive manner, enhancing the representation ability of the features. Experimental results on three public datasets of LEVIR-CD, WHU-CD, and DSIFN-CD show that compared to the existing RSICD methods, the proposed Hybrid-MambaCD achieves the state-of-the-art (SOTA) detection performance.**

***Index Terms*— Hybrid Mamba and convolutional neural network (CNN) architecture, iterative global–local feature fusion (IGLFF), region–channel attention mechanism (RCAM), remote sensing image change detection (RSICD).**

I. Introduction

REMOTE sensing image change detection (RSICD) is to identify and locate change areas from multiple remote sensing images taken over different time periods of the same geographic area. It has wide applications in disaster monitoring [1], agricultural management [2], urban planning [3], [4], and other fields. However, RSICD tasks still face various challenges, such as susceptibility to seasonal changes, lighting condition, shadows, and angle variations. The scale of changing areas often varies drastically, and the background environments are complex and diverse.

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In recent years, RSICD methods based on convolutional neural network (CNN) and Transformer have made significant advances [5]. CNN can effectively extract local features through convolution operations [6]. It has demonstrated exceptional performance and exhibited broad application prospects in various applications, such as hyperspectral image super-resolution (HISR) [7], [8], [9], [10].

Transformer can capture long-range dependency through a self-attention mechanism [11], [12]. However, these methods have certain limitations. CNN is good at extracting local details but the consecutive downsampling operations result in the loss of global context information. Conversely, Transformer can address this issue but has a large number of model parameters and high computational complexity, and it struggles to capture local detail information. Therefore, researchers have proposed hybrid frameworks that combine CNN and Transformer [13], [14], [15], fully leveraging the advantages of CNN for local feature extraction and Transformer for long-range dependency modeling. These methods achieve better change detection performance compared to pure CNN or Transformer. However, these hybrid architectures usually have large model parameters and, simultaneously, incomplete detection of change regions, and particularly missed detection for scattered small objects also often occur.

Recent studies have demonstrated the effectiveness of state space models (SSMs) [16] on long-sequence modeling. Among them, Mamba [17] is the most representative one. Mamba also has powerful long-range dependency modeling capabilities, but its computational complexity grows linearly or near linearly with the input sequence, compared to the quadratic complexity of Transformer. The up-to-date research works on Mamba, such as Vim [18] and VMamba [19], have successfully been applied to visual tasks, achieving better performance than Transformer.

RSM-CD [20] is the first to apply Mamba to RSICD tasks, introducing an omnidirectional selective scan module (OSSM) that selectively scans remote sensing images in multiple directions to achieve long-range context modeling. The research results show that it could achieve better detection performance than Transformer-based methods while maintaining linear computational complexity. However, it is weak in capturing local features, leading to frequent missed and incorrect detections of small, scattered targets.

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To address this issue, in this article, a hybrid Mamba and CNN architecture is proposed for RSICD tasks, named Hybrid-MambaCD. It combines CNN with Mamba to mitigate the local feature representation problems of RSM-CD. Hybrid-MambaCD consists of two branches: one branch uses Mamba to extract global context information, while the other branch employs CNN to extract local detail information. A progressive multiscale feature fusion strategy is then designed to fuse the global and local features, and the fused features are used for change detection.

The main contributions of this article are as follows.

1) A hybrid Mamba–CNN architecture is constructed for the RSICD tasks. Compared to the existing CNN+Transformer counterparts, it obtains more representative global–local features with lower model parameters.

2) Region–Channel attention mechanism (RCAM) is designed to enhance the important CNN features from both the region and channel dimensions, enabling the network to focus more on change regions while suppressing background interference.

3) An iterative global–local feature fusion (IGLFF) strategy is proposed, which progressively fuses the global and local features across multiple scales, enhancing feature representation capabilities and improving the detection performance.

Experimental results on three public datasets of LEVIR-CD, WHU-CD, and DSIFN-CD demonstrate that the proposed Hybrid-MambaCD outperforms the existing RSICD methods, achieving superior detection performance.

In the rest of this article, after reviewing related works in Section II, a detailed description of the proposed Hybrid-MambaCD is provided in Section III. Section IV reports the experimental results and analysis. Finally, the conclusions are drawn in Section V.

II. Related Works

Current RSICD methods can be roughly divided into three categories: CNN-based methods, Transformer-based methods, and Mamba-based methods. Sections II-A–II-C will review the research advances on these three kinds of methods.

A. CNN-Based RSICD Methods

Daudt et al. [21] proposed the fully convolutional early fusion (FC-EF) method, which was the first to apply CNNs to RSICD tasks. This network adopts the U-Net architecture to enable end-to-end detection and significantly outperforms the traditional methods. They further proposed two methods based on the Siamese architecture but with different skip connection strategies: fully convolutional Siamese-concatenation (FC-Siam-conc) and fully convolutional Siamese-difference (FC-Siam-diff). These approaches, however, have some limitations, such as insufficient discriminative features, leading to the detection results with incomplete change regions and irregular boundaries. To address these issues, some researchers introduced various attention mechanisms into CNN, generating different RSICD methods, including DTCDSCN [22], STANet [23], SNUNet-CD [24], DDCNN [25], and HCGMNet [26].

CNN utilizes multiple convolutional kernels with fixed size to extract local information, which makes CNN-based RSICD methods struggle to capture long-range dependency. Consequently, these methods often exhibit missed or incorrect detections, particularly for sparse and scattered small targets.

B. Transformer-Based RSICD Methods

Transformer is good at modeling long-range dependency, so it is applied to RSICD tasks, achieving a superior detection performance compared to CNN-based methods. Zhang et al. [27] proposed SwinSUNet, which embeds the Transformer architecture into a Siamese U-shaped framework, improving the network's ability to extract global contextual information. Hu et al. [28] proposed a global multihead interactive self-attention change detection network, which outperforms the state-of-the-art (SOTA) methods with higher accuracy and efficiency. Other notable works, such as ChangeFormer [29] and M-Swin [30], have also achieved excellent detection results.

Compared to CNN-based methods, Transformer-based methods often can achieve better performance. However, these methods suffer from quadratic growth in computational complexity, requiring much more computational resources. In addition, it is hard for Transformer to effectively capture local detail information, which is crucial for RSICD tasks, leading to poor performance on the remote sensing images with complex ground scenes.

To address these issues of both CNN- and Transformer-based methods, researchers have proposed hybrid architectures to integrate the advantages of both. Chen et al. [13] used a bi-temporal image Transformer (BIT) for RSICD tasks, representing bi-temporal images as tokens and using Transformer to model global context in a compact token-based spatiotemporal space, achieving higher performance than previous methods. Other methods, including H-TransCD [31], TransUNetCD [32], and DBFGAN [15], have also demonstrated superior performance, though at the cost of increased model complexity.

C. Mamba-Based RSICD Methods

Recently, Mamba has become a research focus due to its linear complexity and its ability to achieve performance comparable to or even better than Transformer. Researchers have explored its potential in RSICD tasks. In addition to RSM-CD [20], Chen et al. [33] proposed ChangeMamba, which utilizes VMamba as the backbone network and introduces three spatiotemporal relationship modeling mechanisms into the decoder, effectively enabling the spatiotemporal interaction of features to capture precise change information. These works demonstrate the great potential of Mamba in RSICD tasks. However, while Mamba possesses powerful capabilities for modeling long-range dependency, it often overlooks local features, which presents certain limitations for RSICD tasks.

In this article, a hybrid Mamba–CNN architecture that combines CNN and Mamba is proposed to address the limitations of pure Mamba-based RSICD methods in local feature extraction. It can achieve a superior CD performance with lower model complexity.

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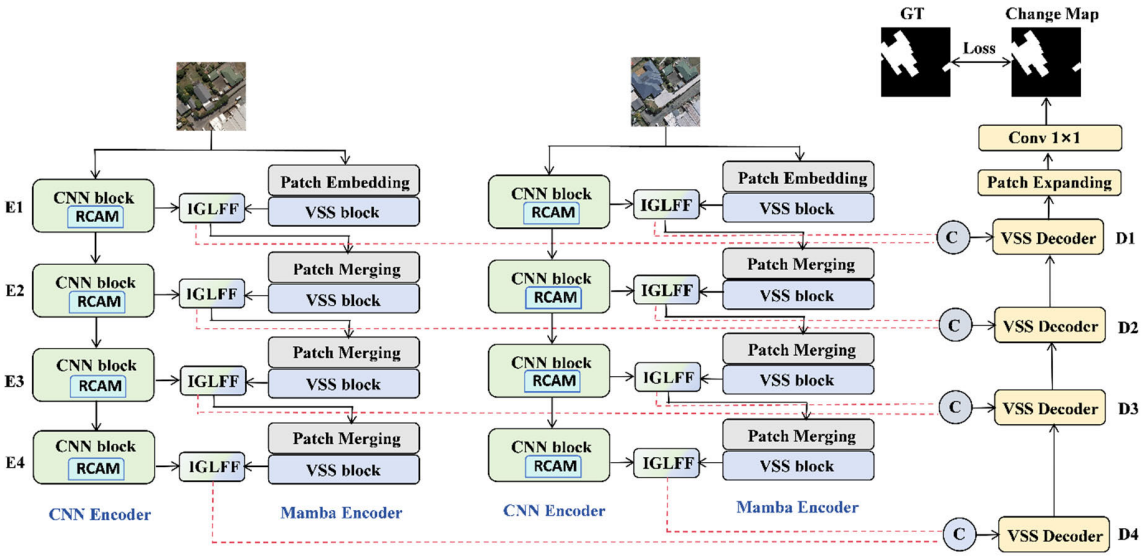


Fig. 1. Overall network architecture of the proposed Hybrid-MambaCD.

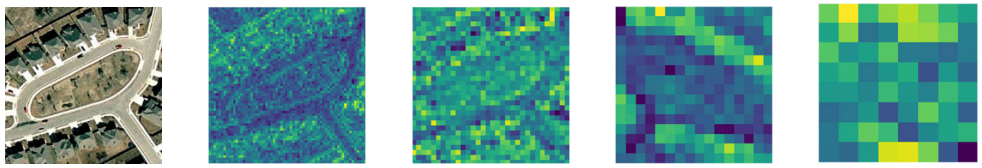


Fig. 2. Visualization of multiscale features.

III. Proposed Method

The proposed Hybrid-MambaCD architecture is illustrated in Fig. 1. It primarily consists of two branches: the Mamba global branch and the ResNet18 local branch. First, the input images are fed into two parallel branches that extract multiscale global and local features. Both branches are composed of four stages (E1∼E4) and sequentially obtain multiscale features with resolution of $(H/4) \times (W/4)$, $(H/8) \times (W/8)$, $(H/16) \times (W/16)$, and $(H/32) \times (W/32)$, respectively, through downsampling operations. The output feature maps of each stage are visualized in Fig. 2. As shown in the figures, the shallow layers of the network contain more low-level visual information such as edges and textures, while the deeper layers contain more global semantic information.

Subsequently, the features extracted from the two branches are fed into the feature fusion module to integrate global and local information. The dual-temporal fused features at different scales are concatenated, yielding four fused features. These four scale features are then input into the Mamba decoder with skip connections. The decoder processes them at four stages (D1∼D4) to generate a binary change map.

To enhance the representation capabilities of features and suppress background interference, RCAM is designed to enhance the local features extracted from ResNet18. Furthermore, to facilitate the fusion of features from the two branches, the IGLFF strategy is then proposed. Sections III-A–III-D will provide detailed implementations of the Mamba encoder and decoder, as well as RCAM and IGLFF.

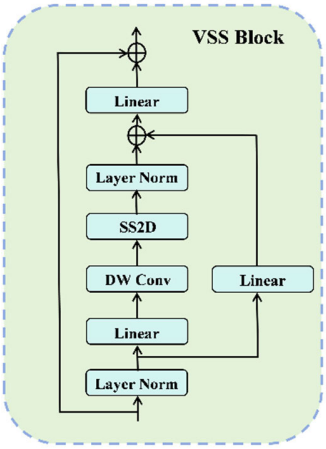


Fig. 3. Architecture of VSS block.

A. Mamba Encoder

Mamba encoder primarily consists of multiple VSS blocks [34], as illustrated in Fig. 3. It can be viewed as a residual network with skip connections, where one branch uses 3 × 3 depth-wise convolutional layers for feature extraction, while the other branch comprises linear mappings and activation

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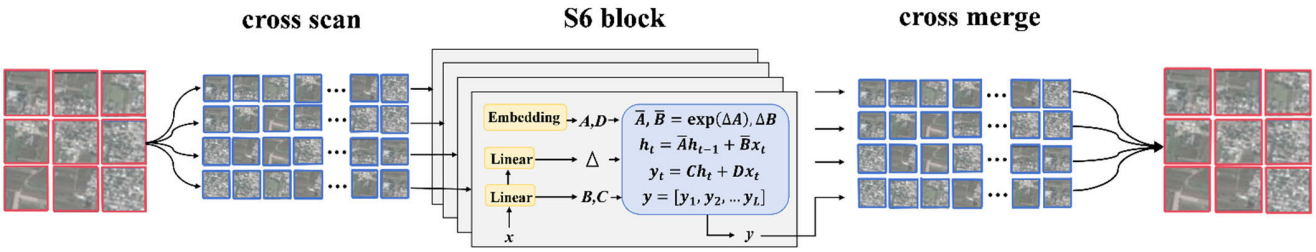


Fig. 4. Illustration of the SS2D operation [19].

layers. The core computational unit of the VSS block is the 2D-selective-scan (SS2D) [19], which enables the processing of 2-D images. SS2D unfolds image patches in four directions, creating four distinct sequences. It then scans in four different directions, that is, from the top left to the bottom right, from the bottom right to the top left, from the top right to the bottom left, and from the bottom left to the top right. This scan way ensures that each pixel integrates information from all other pixels in different directions. Finally, the output results are merged to form a complete 2-D feature map. The specific process is shown in Fig. 4. Assuming that the input data are denoted as x, the output of SS2D can be represented as $\bar{x}$

$$x_v = \text{expand}(x, v) \tag{1}$$
$$\bar{x}_v = S6(x_v) \tag{2}$$
$$\bar{x} = \text{merge}\left(\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4\right) \tag{3}$$

where $v \in V = \{1, 2, 3, 4\}$ represents the four different scanning directions and expand$(\cdot)$ and merge$(\cdot)$ represent the scanning unfolding and scanning merging operations, respectively. In (2), the selective scanning spatial state sequence model (S6) is the core SSM operator of the VSS block. It enables each element in the 1-D array to interact with any previously scanned sample through compressed hidden states.

Mamba encoder consists of four stages (E1∼E4). The first stage, E1, includes patch embedding and VSS block, while the subsequent three stages comprise patch merging and VSS blocks, as shown in Fig. 1. Initially, the patch embedding divides the image into nonoverlapping patches, followed by the VSS block to extract the features of image. After passing through the four stages, the features with four different scales are obtained, denoted as F_M^i, $i = 1, 2, 3, 4$.

B. Region–Channel Attention Mechanism

One of the challenges faced in the RSICD tasks is complex and diverse ground scenes. To mitigate their interference on the detection performance, RCAM is proposed in this article. It enhances important feature channels and suppresses the features of background regions from both channel and spatial dimensions, allowing the network to focus more on the change regions, and thus alleviating the issues of missed and false detections for dispersed and small targets.

RCAM consists of two components: region-aware attention (RA) and channel attention (CA). RA is designed to enhance the features of change regions while suppressing those of background areas. CA, on the other hand, focuses on enhancing

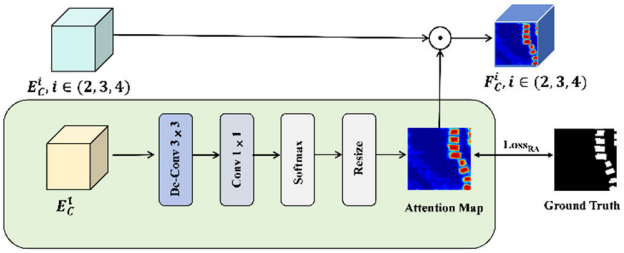


Fig. 5. Architecture of change RA.

important feature channels while suppressing irrelevant ones. The implementation details of these two attention mechanisms are described as follows.

The network structure of RA is illustrated in Fig. 5, and the specific process is as follows.

First, for the input image $X \in R^{256\times256\times3}$, the output feature maps from the first layer of CNN branch are denoted as $E_C^1 \in R^{64\times64\times64}$. The deconvolution is then applied to E_C^1, obtaining D_m, which has a size of 256 × 256. Next, a 1 × 1 convolution operation is applied to D_m for dimensionality reduction, enabling information exchange between channels without altering the size of dimensions of the feature maps, resulting in D_n. Subsequently, the softmax activation function is used to generate the regional attention map $\alpha \in R^{256\times256\times1}$. α is resized through convolution operation to match the scale of E_C^i, resulting in the regional attention maps α_i, $i = 2, 3, 4$. Then, use α_i to weight the features E_C^i, $i = 2, 3, 4$, at different scales, and obtain the enhanced features F_C^i, $i = 2, 3, 4$. The entire operation process can be represented by the following formulas:

$$D_m = \text{Deconv}_{3\times3}\left(F_C^1\right) \tag{4}$$
$$D_n = \text{conv}_{1\times1}(D_m) \tag{5}$$
$$\alpha = \text{softmax}(D_n) \tag{6}$$
$$\alpha_i = \text{resize}(\alpha), \quad i = 2, 3, 4 \tag{7}$$
$$F_C^i = \alpha_i \cdot E_C^i, \quad i = 2, 3, 4. \tag{8}$$

Due to the drastic changes and different shapes of contours in remote sensing images, to ensure more accurate detection of change regions, in this article, binary cross-entropy (BCE) loss function is adopted to constrain the generation process of the regional attention map. The BCE loss measures the distance between the output probabilities and the ground truth (GT), making it particularly suitable for binary classification problems, such as pixel-level classification (change or no change)

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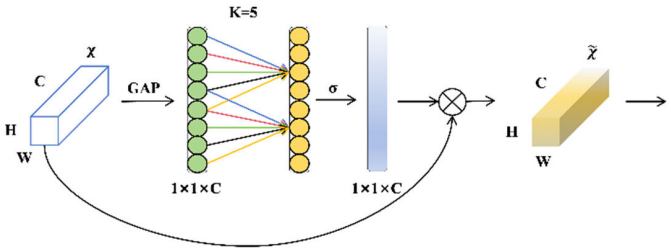


Fig. 6. Architecture of ECA.

in RSICD. In addition, the BCE loss is computationally simple and efficient, serving as a complementary addition to the primary loss functions. This allows the network to better focus on change regions while suppressing irrelevant background noise. The formula of BCE loss is as follows:

$$\text{Loss}_{\text{RA}} = -\frac{1}{n}\sum(y_n \times lnx_n + (1-y_n) \times ln(1-x_n)) \ldots \tag{9}$$

where y_n is the GT and x_n is the predicted output map.

CA is used to further enhance important feature channels and suppress irrelevant ones. In this article, the multiscale features enhanced by RA are further enhanced through the efficient CA (ECA) mechanism [35]. ECA is a CA mechanism proposed by Wang et al. [35], and its structure is illustrated in Fig. 6. Compared to other CA mechanisms, ECA avoids complex dimensionality reduction and expansion operations, making it efficient, lightweight, and easy to insert. ECA has demonstrated excellent performance in various tasks, including image classification and segmentation. Therefore, in this article, ECA is used for feature enhancement. The output features of ECA are denoted as $\hat{F}_C^i$, $i = 2, 3, 4$.

C. Iterative Global–Local Feature Fusion

The Mamba and CNN branches extract global and local features, respectively, each containing different information. Fusing these features to generate more representative global–local features is crucial for detection performance improvement. In this article, the IGLFF strategy is proposed, whose structure is illustrated in Fig. 7.

The input of IGLFF consists of three components: global features F_M^i, local features $\hat{F}_C^i$ enhanced by RCAM, and features G_{i-1} fused from the previous stage. The outputs are the fused features of the ith scale, denoted as $G_i, i = 2, 3, 4$. First, features G_{i-1} are downsampled to obtain features $\hat{G}_{i-1}$, and F_M^i, $\hat{F}_C^i$, and $\hat{G}_{i-1}$ are concatenated together to form G_i'. The spatial attention (SA) mechanism is employed to enhance the key information in the global features while suppressing irrelevant ones, resulting in the enhanced features $\hat{F}_M^i$. Then, $\hat{F}_C^i$, $\hat{F}_M^i$, and G_i' are concatenated together to form G_i''. Finally, the global average pooling (GAP) operation is applied to obtain the fused features, which contain both global and local information.

Considering the varying contributions of different features to detection performance, in this article, the weights of $\hat{F}_M^i$, $\hat{F}_C^i$, and G_i' are determined in an adaptive manner through learning. Specifically, a fully connected (FC) layer is used to generate different weights for each of the three features, and each feature channel is weighted to obtain the fused features. At the end of IGLFF, a residual convolutional multilayer perceptron (RCMLP) is exploited to prevent the issues of gradient explosion/vanishing. The entire process is described by (10)–(19)

$$\text{SA}(x) = \sigma\left(f_{7\times7}\left(\text{Concat}\left[\text{AvgPool}(x), \text{MaxPool}(x)\right]\right)\right) \tag{10}$$

$$\hat{F}_M^i = \text{SA}\left(F_M^i\right) \otimes F_M^i \tag{11}$$

$$\text{RCMLP}(x) = f_{1\times1}(f_{1\times1}(f_{3\times3}(LN(x)) + LN(x))) \tag{12}$$

where σ represents the sigmoid activation function, $f_{7\times7}$ denotes a 7×7 convolution operation, and SA refers to the SA mechanism operation

$$\hat{G}_{i-1} = \text{AvgPool}(f_{1\times1}(G_{i-1})) \tag{13}$$

$$G_i' = f_{3\times3}\left(\text{Concat}\left[F_M^i, \hat{F}_C^i, \hat{G}_{i-1}\right]\right) \tag{14}$$

$$G_i'' = f_{3\times3}\left(\text{Concat}\left[\hat{F}_M^i, \hat{F}_C^i, G_i'\right]\right) \tag{15}$$

$$\tilde{F}_M^i = \text{Softmax}\left(\text{FC}_1\left(\text{GAP}\left(G_i''\right)\right)\right) \otimes \hat{F}_M^i \tag{16}$$

$$\tilde{G}_i = \text{Softmax}\left(\text{FC}_2\left(\text{GAP}\left(G_i''\right)\right)\right) \otimes G_i' \tag{17}$$

$$\tilde{F}_C^i = \text{Softmax}\left(\text{FC}_3\left(\text{GAP}\left(G_i''\right)\right)\right) \otimes \hat{F}_C^i \tag{18}$$

$$G_i = \text{RCMLP}\left(\text{Sum}\left[\tilde{F}_M^i, \tilde{G}_i, \tilde{F}_C^i\right]\right) + \hat{G}_{i-1} \tag{19}$$

where $\otimes$ represents the element-wise multiplication operation, GAP denotes the GAP operation, and FC stands for the FC layer.

D. Mamba Decoder

In this article, a lightweight VSS decoder is employed, aiming to achieve good detection performance with fewer model parameters. The decoder mainly consists of patch expanding blocks and VSS blocks. The first four patch expanding blocks perform an upsampling operation with a factor of 2, while the last patch expanding block performs an upsampling operation with a factor of 4. It performs skip connections with the bi-temporal fused features obtained from the encoder, enhancing the representative power of the features.

E. Loss Function

In this article, focal loss and Dice loss are combined together to optimize the network training. Focal loss is a loss function designed to handle class imbalance problem. Dice loss is commonly used for pixel-level image segmentation tasks and is better suited than standard cross-entropy loss for dealing with imbalanced classes in segmentation tasks. By combining the two losses, the model can effectively address the issue of class imbalance between change and no-change regions in the RSICD tasks. The total loss is expressed as follows:

$$\text{total_loss} = \text{Dice_loss} + \text{Focal_loss} + \text{Loss}_{\text{RA}}. \tag{20}$$

Loss_{RA} is shown as (9), and the mathematical formulas for Dice loss and focal loss are as follows:

$$\text{Dice_loss} = 1 - \text{Dice} = 1 - \frac{2|X \cap Y|}{|X| + |Y|} \tag{21}$$

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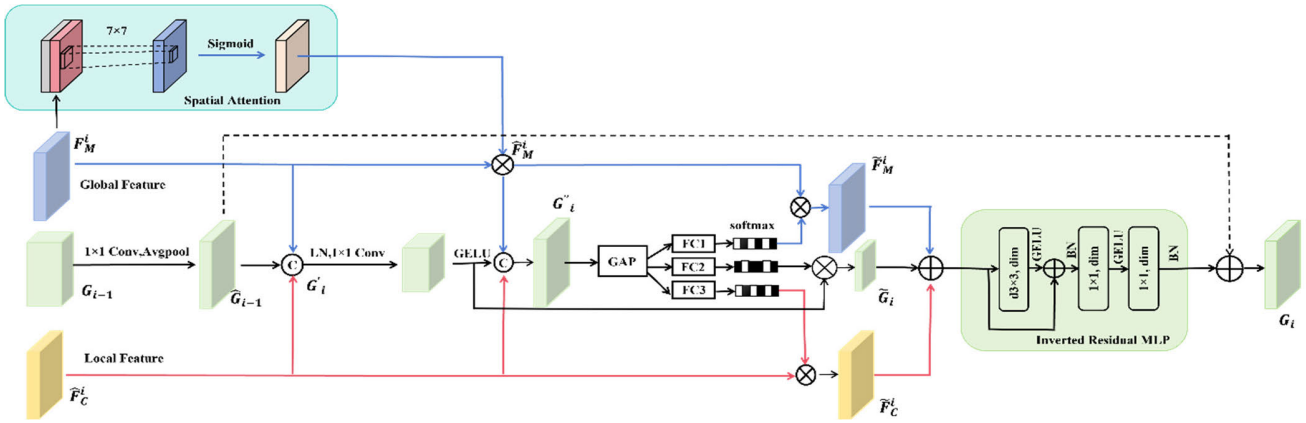


Fig. 7. Architecture of IGLFF.

$$\text{Focal_loss}(p_t) = -\alpha_t(1-p_t)^{\gamma}\log(p_t) \tag{22}$$

where X represents the pixel labels of GT and Y denotes the pixel classes of the predicted change map. The parameter α_t serves as a balancing factor, and $(1-p_t)^{\gamma}$ serves as an adjustment mechanism.

IV. Experiment Results and Analysis

To evaluate the effectiveness of the proposed Hybrid-MambaCD, we conducted experiments and assessments on three publicly available RSICD datasets. The experimental results and analysis are presented as follows.

A. Datasets and Evaluation Metrics

1) Datasets: The three public datasets used in the experiments are LEVIR-CD [23], WHU-CD [36], and DSIFN-CD [37].

LEVIR-CD is a large-scale public dataset for building change detection, containing 637 pairs of high-resolution (0.5 m) Google Earth images, each sized at 1024 × 1024 pixels, covering 20 different regions from 2002 to 2018.

WHU-CD focuses on building change detection and consists of a pair of high-resolution (0.075 m) aerial images, measuring 32 507 × 15 354 pixels, covering areas in Christchurch, New Zealand, from 2012 to 2016.

DSIFN-CD encompasses changes in various land cover categories, derived from six pairs of high-resolution (2 m) satellite images.

In the experiments, paired image samples from the three datasets were uniformly cropped into nonoverlapping patches of size 256 × 256. The resulting sample pairs were used in experiments. All three datasets were divided into training, validation, and test sets according to the default settings.

2) Performance Evaluation Metrics: In this article, five commonly used CD performance evaluation metrics, i.e., precision (Pre), recall (Rec), $F1$-score ($F1$), intersection over union (IoU), and overall accuracy (OA), were adopted. False positives (FPs) and false negatives (FNs) refer to pixels incorrectly classified as changed and unchanged, respectively. True positives (TPs) represent correctly detected changed pixels, while true negatives (TNs) represent correctly detected unchanged pixels. The five metrics can be defined as follows.

a) Precision: It measures the proportion of TP samples among all samples classified as positive. The formula is as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP}+\text{FP}}. \tag{23}$$

b) Rec: It measures how many of all positive samples are classified as positive. The formula is as follows:

$$\text{Rec} = \frac{\text{TP}}{\text{TP}+\text{FN}}. \tag{24}$$

c) F1-score: It is a metric that combines both precision and recall, providing a more comprehensive assessment of detection performance. The formula is as follows:

$$F1 = \frac{2*\text{Pre}*\text{Rec}}{\text{Pre}+\text{Rec}}. \tag{25}$$

d) Intersection over union: It evaluates the overlap between the detection results and the GT. The formula is as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}}. \tag{26}$$

e) Overall accuracy: It measures the proportion of correctly classified pixels in the detection results relative to the total number of pixels in the entire image. The formula is as follows:

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FN}+\text{TN}+\text{FP}}. \tag{27}$$

B. Parameter Settings

The proposed architecture was constructed, trained, and tested on the PyTorch platform. The hardware configuration was as follows: Intel Core i5-7500 CPU at 3.40 GHz, and graphics card is NVIDIA TITAN Xp. During training, data augmentation methods, such as random flipping ($p = 0.5$), transposition ($p = 0.5$), and swapping dual temporal images ($p = 0.5$), were used. The AdamW optimizer was adopted with an initial learning rate of $1e^{-5}$ and a weight decay of 0.001.

The learning rate adjustment strategy is as follows; if the $F1$-score on the validation set does not improve over a span

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TABLE I
QUANTITATIVE COMPARISONS ON THE LEVIR-CD DATASET

Method	Year	Architecture	Pre	Rec	F1	IoU	OA
FC-EF [21]	2018	CNN	86.91	80.17	83.40	71.53	98.39
FC-Siam-Di [21]	2018	CNN	89.53	83.31	86.31	75.92	98.67
FC-Siam-Conc [21]	2018	CNN	91.99	76.77	83.69	71.96	98.49
DTCDSCN [22]	2020	CNN	88.53	86.83	87.67	78.05	98.77
STANet [23]	2020	CNN	83.81	**91.00**	87.26	77.40	98.66
IFNet [37]	2020	CNN	**94.02**	82.93	88.13	78.77	98.87
SNUNet [24]	2021	CNN	89.18	87.17	88.16	78.83	98.82
BIT [13]	2021	CNN+Transformer	89.24	89.37	89.31	80.68	98.92
H-TransCD [31]	2022	CNN+Transformer	91.45	88.72	90.06	81.92	99.00
ChangeFormer [29]	2022	Transformer	92.05	88.80	90.40	82.48	99.04
DMINet [38]	2023	Transformer	92.52	89.95	90.71	82.99	**99.07**
RSM-CD [20]	2024	Mamba	**92.52**	89.73	**91.10**	**83.66**	/
Ours	2024	Mamba+CNN	92.20	**90.78**	**91.48**	**84.31**	**99.13**

TABLE II
QUANTITATIVE COMPARISONS ON THE WHU-CD DATASET

Method	Year	Architecture	Pre	Rec	F1	IoU	OA
FC-EF [21]	2018	CNN	71.63	67.25	69.37	53.11	97.61
FC-Siam-Di [21]	2018	CNN	47.33	77.66	58.81	41.66	95.63
FC-Siam-Conc [21]	2018	CNN	60.88	73.58	66.63	49.95	97.04
DTCDSCN [22]	2020	CNN	63.92	82.30	71.95	56.19	97.42
STANet [23]	2020	CNN	79.37	85.50	82.32	69.95	98.52
IFNet [37]	2020	CNN	**96.91**	73.19	83.40	71.52	98.83
SNUNet [24]	2021	CNN	85.60	81.49	83.50	71.67	98.71
BIT [13]	2021	CNN+Transformer	86.64	81.48	83.98	72.39	98.75
H-TransCD [31]	2022	CNN+ Transformer	93.85	88.73	91.22	83.85	**99.24**
ChangeFormer [29]	2022	Transformer	91.83	88.02	89.88	81.63	99.12
DMINet [38]	2023	Transformer	93.84	86.25	88.69	79.68	98.97
RSM-CD [20]	2024	Mamba	93.37	**90.42**	**91.87**	**84.96**	/
Ours	2024	Mamba+CNN	**95.51**	**92.52**	**93.99**	**88.66**	**99.53**

of ten epochs, the learning rate is reduced by a factor of 0.1. The input image size is normalized to 256 × 256, the batch size is set to 8, and the total number of epochs is 200.

C. Comparison With SOTA RSICD Methods

We compared the proposed Hybrid-MambaCD with 12 SOTA methods on the LEVIR-CD, WHU-CD, and DSIFN-CD datasets. The comparison methods include FC-EF [20], FC-Siam-Di [20], FC-Siam-Conc [20], DTCDSCN [22], STANet [23], IFNet [38], SNUNet [24], BIT [13], H-TransCD [31], ChangeFormer [29], DMINet [38], and RSM-CD [20]. Among these, FC-EF, FC-Siam-Di, FC-Siam-Conc, DTCDSCN, STANet, IFNet, and SNUNet are CNN-based methods. ChangeFormer and DMINet are Transformer-based methods, and H-TransCD and BIT are hybrid CNN-Transformer methods. RSM-CD is Mamba-based method.

The comparison results on the three datasets using different methods are shown in Tables I∼III, in which the optimal and suboptimal results are marked in red and blue, respectively. The data for other methods other than ours are cited from the literature.

The comparison results reveal that the proposed Hybrid-MambaCD achieves optimal or suboptimal metrics in most cases. Next, we will provide a detailed analysis of the experimental results on each dataset.

1) *LEVIR-CD Dataset*: Except for the precision metric, the proposed Hybrid-MambaCD achieves the optimal results in terms of $F1$, IoU, and OA metrics, while recall has achieved a suboptimal result. Compared to the suboptimal method RSM-CD, the proposed Hybrid-MambaCD improves the recall, $F1$, and IoU metrics by 1.05%, 0.38%, and 0.65%, respectively. Although its precision metric is slightly lower, it is only 0.32% lower than that of RSM-CD.
2) *WHU-CD Dataset:* The proposed Hybrid-MambaCD achieves optimal results in terms of recall, $F1$, IoU, and OA metrics, while the precision metric reaches a suboptimal result. Compared to the suboptimal method RSM-CD, the proposed Hybrid-MambaCD improves the precision, recall, $F1$, and IoU metrics by 2.14%, 2.10%, 2.12%, and 3.70%, respectively.
3) *DSIFN-CD Dataset*: The proposed Hybrid-MambaCD achieves optimal results on five metrics. Compared to the suboptimal method ChangeFormer, the proposed Hybrid-MambaCD improves these five metrics by

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TABLE III
QUANTITATIVE COMPARISONS ON THE DSIFN-CD DATASET

Method	Year	Architecture	Pre	Rec	F1	IoU	OA
FC-EF [21]	2018	CNN	72.61	52.73	61.09	43.98	88.59
FC-Siam-Di [21]	2018	CNN	59.67	65.71	62.54	45.50	86.63
FC-Siam-Conc [21]	2018	CNN	66.45	54.21	59.71	42.56	87.57
DTCDSCN [22]	2020	CNN	53.87	77.99	63.72	46.76	84.91
STANet [23]	2020	CNN	67.71	61.68	64.56	47.66	88.49
IFNet [37]	2020	CNN	67.86	53.94	60.10	42.96	87.83
SNUNet [24]	2021	CNN	60.60	72.89	66.18	49.45	87.34
BIT [13]	2021	CNN+Transformer	68.36	70.18	69.26	52.97	89.41
ChangeFormer [29]	2022	Transformer	**88.48**	**84.94**	**86.67**	**76.48**	**95.56**
Ours	2024	Mamba	**96.16**	**95.88**	**96.02**	**92.34**	**98.66**

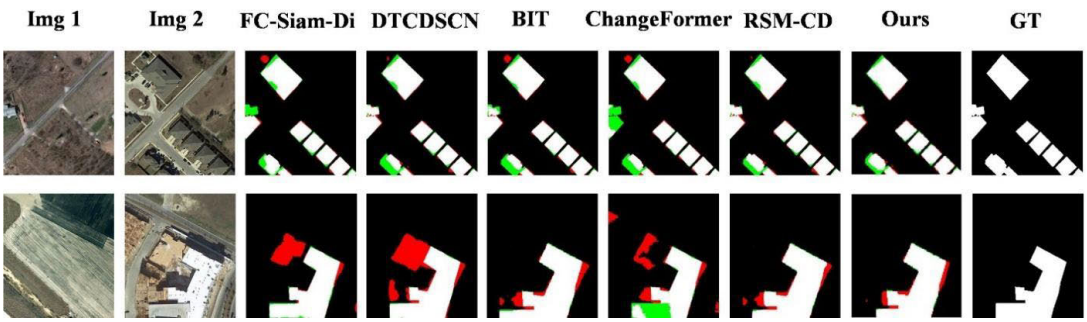


Fig. 8. Visualized results of different methods on the LEVIR-CD dataset.

TABLE IV
COMPARISON OF COMPUTATIONAL COMPLEXITY AND MODEL PARAMETERS WITH OTHER METHODS

Method	Params. (M)	FLOPs (G)	Pre	Rec	F1	IoU	OA
STANet [23]	16.93	**6.58**	83.81	**91.00**	87.26	77.40	98.66
SNUNet [24]	**12.03**	27.44	89.18	87.17	88.16	78.83	98.82
TransUNetCD [32]	95.49	27.23	92.43	89.82	91.11	83.67	/
H-TransCD [31]	183.83	67.58	91.45	88.72	90.06	81.92	99.00
ChangeFormer [29]	41.03	101.57	92.05	88.80	90.40	82.48	99.04
RSM-CD [20]	27.90	15.70	**92.52**	89.73	91.10	83.66	/
Ours	41.51	65.07	92.20	90.78	**91.48**	**84.31**	**99.13**

7.68%, 10.94%, 9.35%, 15.86%, and 3.10%, achieving the SOTA performance on this dataset.

To visually demonstrate the detection performance of Hybrid-MambaCD, we visualized the detection results of different methods. Fig. 8 presents an example from the LEVIR-CD dataset. The first and second columns represent the dual-temporal images, while the last column shows GT. The detection results are compared with the methods of FC-Siam-Di, DTCDSCN, BIT, ChangeFormer, and RSM-CD. The results are highlighted in different colors: white for TPs, black for TNs, red for FPs, and green for FNs. From the comparison, it can be observed that in the first row, other methods often miss some subtle change regions, while Hybrid-MambaCD successfully detects them, reducing both the missed detection rate and instances of "false changes." The second row shows that the changes detected by Hybrid-MambaCD have more complete and accurate edges. In areas without changes, other methods incorrectly detected the changes, Hybrid-MambaCD effectively reduced the false detections.

Overall, these experimental results comprehensively validate the effectiveness of the proposed Hybrid-MambaCD, demonstrating its capability to significantly decrease false and missed detections in scattered, subtle change regions, yielding better detection performance.

D. Comparison of Computational Complexity and Model Parameters

To compare the model parameters and computational complexity of the proposed Hybrid-MambaCD, we evaluated it alongside CNN-based methods such as STANET and SNUNet, Transformer-based methods of ChangeFormer, and hybrid CNN+Transformer methods such as H-TransCD and TransUNetCD. The results are shown in Table IV, where

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TABLE V
TRAINGTIME AND INFERENCE TIME

Dataset	Training time	Inference time
LEVIR-CD	66 h	81 s
WHU-CD	56 h	29 s
DSIFN-CD	126 h	18 s

TABLE VI
ABLATION EXPERIMENTAL RESULTS ON THE LEVIR-CD DATASET

Baseline	RCAM	IGLFF	Pre	Rec	F1	IoU	OA
√			91.65	89.74	90.68	82.96	99.06
√	√		92.09	90.33	91.20	83.83	99.11
√		√	91.70	90.12	90.90	83.32	99.07
√	√	√	**92.20**	**90.78**	**91.48**	**84.31**	**99.13**

computational complexity is measured in floating-point operations per second (FLOPs). The key findings are as follows.

1) Compared to ChangeFormer, the proposed Hybrid-MambaCD has only 67% of its model parameters, yet its detection performance outperforms in recall, $F1$, IoU, and OA metrics by 1.98%, 1.08%, 1.83%, and 0.09%, respectively.

2) Compared to the TransUNetCD method, the proposed Hybrid-MambaCD has only 43.47% of its parameter scale, while achieving improvements of 0.96%, 0.37%, and 0.64% in recall, $F1$, and IoU metrics, respectively. In comparison to H-TransCD, Hybrid-MambaCD has just 22.58% of its parameters and lower computational complexity, yet its detection performance surpasses H-TransCD in precision, recall, $F1$, IoU, and OA metrics by 0.75%, 2.06%, 1.42%, 2.39%, and 0.13%, respectively.

3) Compared to CNN-based methods, the proposed Hybrid-MambaCD model has a higher parameter scale and computational complexity. However, it achieves significant performance improvements in precision, recall, $F1$, IoU, and OA metrics. Specifically, compared to SNUNet, Hybrid-MambaCD improves these five metrics by 3.02%, 3.61%, 3.32%, 5.48%, and 0.31%.

4) Compared to RSM-CD, the proposed Hybrid-MambaCD has a slightly higher parameter scale and computational complexity. However, it outperforms RSM-CD in recall, $F1$, and IoU metrics by 1.05%, 0.38%, and 0.65%, respectively.

Overall, our proposed Hybrid-MambaCD achieves a superior detection performance without significantly increasing the model parameters and computational complexity.

E. Training Time and Inference Time

The training time and inference time of the proposed method on three datasets are shown in Table V. Both training and inference were conducted on images of size 256 × 256 pixels.

F. Ablation Studies

To validate the effectiveness of each component of the proposed Hybrid-MambaCD, extensive ablation experiments were conducted on the LEVIR-CD dataset.

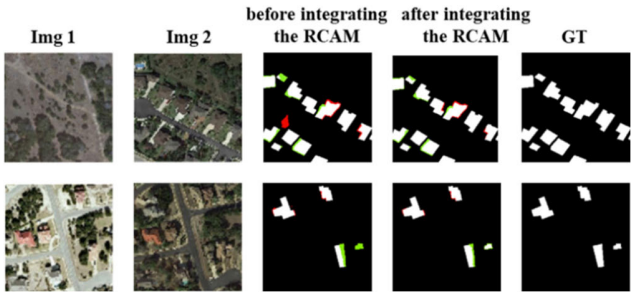


Fig. 9. Visualized results of RCAM on the LEVIR-CD dataset.

1) Effectiveness of Each Component: The results of the ablation experiments on the LEVIR-CD dataset are shown in Table VI. The baseline represents the results obtained by simply concatenating the features from the Mamba branch and the CNN branch, where the CNN branch does not utilize the RCAM.

The experimental results demonstrate that both the RCAM and IGLFF contribute to the improvement of detection performance. Among them, RCAM contributes more than IGLFF, particularly on the recall metric, which improves by 0.59% compared to the baseline. This improvement is attributed to the RCAM's ability to enhance the important features while suppressing the background interference, thereby enabling the network to focus more on the change areas, especially scattered small targets. With the addition of IGLFF, further improvements can be observed across all metrics. The reason is that IGLFF can better integrate global and local features across multiple scales, resulting in a more representative feature representation that boosts the detection performance.

We specifically conducted a visual analysis of the RCAM, as shown in Fig. 9.

The first and second columns represent the bi-temporal images, while the third and fourth columns show the detection results before and after incorporating the RCAM. The fifth column represents GT. Detection results are highlighted in different colors, with white, black, red, and green representing TPs, TNs, FPs, and FNs, respectively. From the first two

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TABLE VII
EFFECT OF RCAM TO DIFFERENT LAYERS IN CNN BRANCH

	Params.(M)	FLOPs(G)	Pre	Rec	F1	IoU	OA
Layer1	47.02	31.11	91.70	90.15	90.92	83.35	99.08
Layer1~2	47.54	32.44	91.82	90.09	90.95	83.40	99.09
Layer1~3	48.00	33.68	92.00	90.08	91.03	83.54	99.10
Layer1~4	51.15	34.86	**92.09**	**90.33**	**91.20**	**83.83**	**99.11**

TABLE VIII
EFFECT OF DIFFERENT MAIN/AUXILIARY BRANCHES

Main Branch	Auxiliary Branch	Pre	Rec	F1	IoU	OA
CNN	Mamba	91.04	89.35	90.18	82.12	99.01
Mamba	CNN	**92.20**	**90.78**	**91.48**	**84.31**	**99.13**

TABLE IX
EFFECT OF DIFFERENT MODELS OF FUSION

Fusion Method	Pre	Rec	F1	IoU	OA
Element-Wise Addition	91.76	89.87	90.81	83.16	99.09
Multiplication	92.01	90.02	91.00	83.49	99.10
Concatenation	92.00	90.08	91.03	83.54	99.10
Adaptive Fusion of Global and Local Branches	91.89	90.34	91.11	83.67	99.10
IGLFF	**92.20**	**90.78**	**91.48**	**84.31**	**99.13**

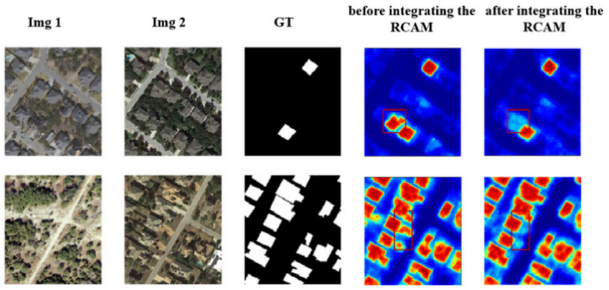


Fig. 10. Heat maps of RCAM on the LEVIR-CD dataset.

rows of the figure, it can be observed that the scattered and small change targets are undetectable before the inclusion of the RCAM. After its integration, these targets are accurately detected, effectively mitigating the missed detections while significantly reducing "false changes."

In addition, we conducted a heat-map analysis of the RCAM, as shown in Fig. 10. The first and second columns represent the dual-temporal images, while the third column shows GT. The fourth and fifth columns display the changes in heat maps before and after incorporating the RCAM. The red boxes highlight the key change areas for comparison. By comparing the heat maps, it is evident that after integrating the RCAM, the network pays more attention to the change areas, thus improving the detection performance.

2) Impact of RCAM Position Within CNN on Detection Performance: We conducted experiments to investigate the impact of integrating the RCAM at different positions within CNN on the detection performance. The results are shown in Table VII. The table also presents the increase in model parameters resulting from the incorporation of RCAM. It is evident that embedding RCAM at all four layers of the CNN yields the best performance, particularly in terms of the recall metric, which shows significant improvement. Notably, this enhancement in performance occurs without a substantial increase in model parameters. Therefore, we chose to integrate the RCAM across all four layers.

3) Impact of Different Main/Auxiliary Branches on Detection Performance: We conducted comparison experiments using different main/auxiliary branches. The results are shown in Table VIII. The first row indicates that when the main branch adopts CNN and the auxiliary branch adopts Mamba. Features extracted from Mamba are fused with those of CNN branch and, then, used as inputs for the next stage of the CNN branch, while the decoder employs convolutional upsampling operations, corresponding to the CNN. The second row represents the proposed method in this article. Both approaches incorporate RCAM and IGLFF. The experimental results demonstrate that using Mamba as the main branch and CNN as the auxiliary branch yields superior detection performance across all metrics. Therefore, Hybrid-MambaCD adopts the second configuration.

4) Impact of Different Feature Fusion Methods on Detection Performance: We conducted the comparison experiments using different feature fusion methods. The results are shown in Table IX. The comparison methods include element-wise addition, element-wise multiplication, feature concatenation, adaptive fusion of global and local branches (without fusing features from the previous stage), and the proposed IGLFF. The experiments demonstrate that IGLFF achieves the best detection performance. This is because the proposed IGLFF

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TABLE X
EFFECT OF BCE LOSS ON DETECTION PERFORMANCE

	Pre	Rec	F1	IoU	OA
ACE Loss	91.86	90.65	91.28	83.90	99.12
BCE Loss	92.20	90.78	91.48	84.31	99.13

can adaptively assign the weights for three kinds of features according to their representation capabilities, obtaining a more powerful representation, and, consequently, a superior detection performance.

5) Effect of BCE Loss on Detection Performance: We conducted experiments to verify the effect of the BCE loss used in the RCA mechanism on the detection performance, and the experimental results are shown in Table X. In this study, we compared BCE loss with ACE loss. The results indicate that employing BCE loss for deep supervision can improve the change detection performance.

V. CONCLUSION

In this article, Hybrid-MambaCD is proposed for RSICD tasks, which applies Mamba to RSICD tasks to fully leverage Mamba's excellent capabilities for extracting global context information, combined with CNN's local feature extraction ability. This approach enables effective detection of change regions with drastic scale variations, particularly for scattered and small targets, mitigating missed and false detections. The following conclusions can be drawn from the experiments.

1) In RSICD tasks, compared to the CNN+Transformer combination, CNN+Mamba achieves a superior detection performance with a lower model parameter scale.

2) By designing the RCAM to enhance features from both channel and region dimensions, the model focuses more on change areas, significantly improving the recall metrics.

3) IGLFF progressively fuses multiscale global and local features, enhancing the representative capability of the fused features.

Experimental results on three public datasets demonstrate that the proposed Hybrid-MambaCD achieves a superior detection performance, compared to the existing methods, providing a highly promising solution for RSICD tasks. However, the proposed method still has certain limitations. Its model parameters are high, which is not conducive to practical applications. In future work, we will design lightweight networks to achieve a better balance between detection performance and model complexity while also striving to design more effective feature fusion and model learning strategies to further improve the detection performance.

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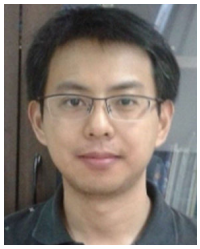
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